

Embracing Artificial Intelligence in Distance Learning Assignments

Enhancing Student Learning and Engagement in the Age of Artificial Intelligence

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NARRATIVE SUMMARY

Educators often struggle with student use of Artificial Intelligence (AI) in the classroom. There has been much debate on how to prevent students from using AI on assignments, how instructors can monitor this usage, and what it could mean for performance indicators when measuring student outcomes.

Teaching proper usage of AI as an educational tool can help students embrace the technology and be more educated consumers in the long run. Instead of preventing students from using this budding technology, there are methods educators can incorporate to explore the benefits of AI while training students in the shortfalls of the service.

Embracing AI as a positive institution in the classroom environment can help students be more critically aware of course topics, sift through more information in a shorter amount of time, and empower them to make more informed decisions. This presentation focuses on practical applications of including AI into distance learning assignments and provides insight as to how students' opinions of the technology changed over the course of a semester.

Introduction

The AI Revolution: AI is reshaping how students learn.

- **Artificial Intelligence (AI) is fundamentally transforming how students learn by personalizing education, automating administrative tasks, and enabling new forms of engagement and feedback.**
- **AI-powered tools such as adaptive learning platforms, intelligent tutoring systems, and generative AI offer tailored instruction, instant answers, and 24/7 support for students.**
 - For example, AI can analyze student performance data to recommend customized learning paths and resources (Zawacki-Richter et al., 2019).
 - AI chatbots and virtual assistants can answer routine questions and free up teachers to focus on higher-level instruction (Luckin et al., 2016).
 - Generative AI models are also being used to help students brainstorm ideas, write drafts, and practice language skills, revolutionizing traditional homework, and research methods (Popenici & Kerr, 2017).
- **While AI enhances access and efficiency, it also raises ethical questions around data privacy, bias, and the role of human educators.**
- **It raises more questions such as how do we navigate responsible use and digital literacy?**

The Role of Instructors: Guide ethical and innovative AI use.

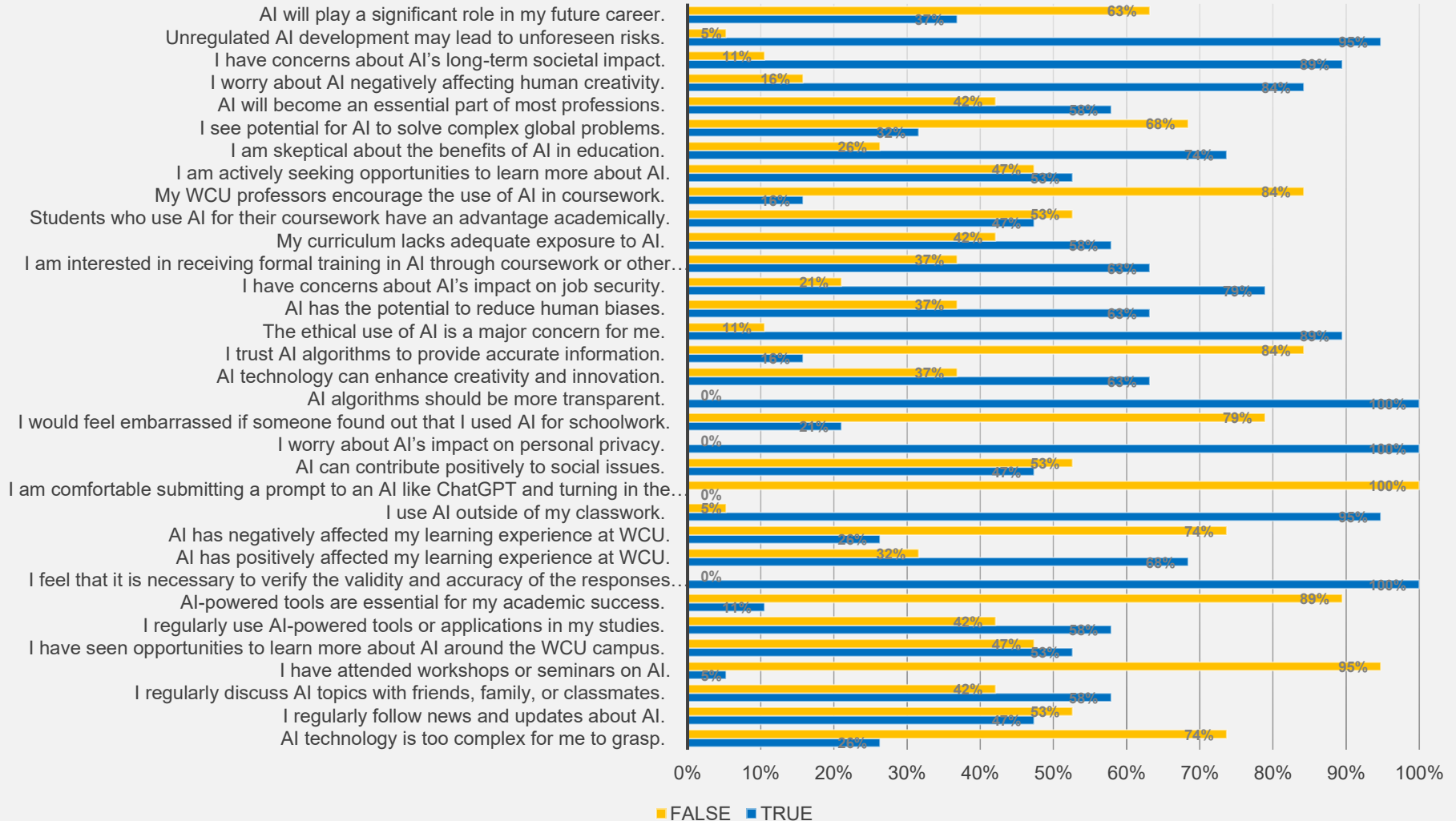
- **As AI tools become more accessible, instructors must help students understand both the benefits and limitations of these technologies, including issues like data privacy, algorithmic bias, plagiarism, and academic integrity (Holmes et al., 2019).**
- **Educators can model responsible AI use by integrating clear policies, promoting transparency about how AI tools work, and cultivating digital literacy skills (Luckin et al., 2016).**
 - They also have the opportunity to design innovative learning activities that harness AI to enhance creativity, problem-solving, and collaboration, while emphasizing that AI should augment, not replace, human judgment (Zawacki-Richter et al., 2019).
- **By guiding students to think critically about AI's societal impact, instructors help prepare graduates to use AI responsibly in their future careers.**

The Chronicle of Higher Education

- “Artificial intelligence and higher education have had a difficult, complex relationship.
- While 86 percent of college students use AI for studies, the technology hasn't been embraced in the same way by college faculty.”
- “Eighty-three percent of college administrators agree or strongly agree that GenAI can improve communication with students, according to a [2024 survey from *The Chronicle*](#).
- Yet, only 35 percent of colleges offer a chatbot for student support.” (McMourtrie, 2023)

AI Survey Data

SURVEY DATA OVERVIEW

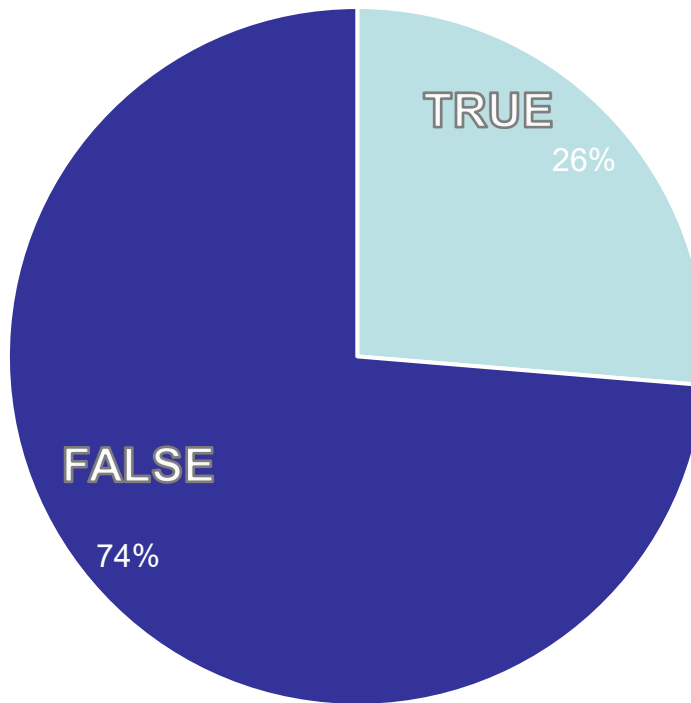


AI technology is too complex for me to grasp.

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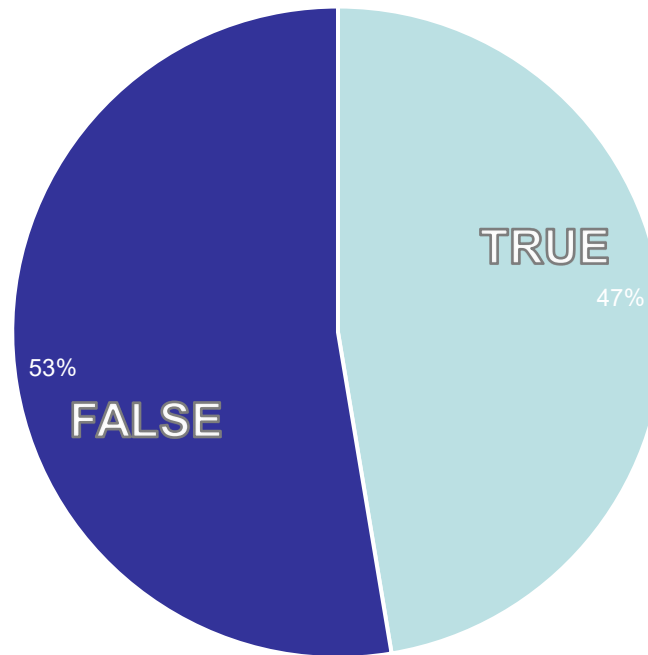
Majors Represented:

- Computer Science
- Nursing
- Criminal Justice
- Engineering
- Communications
- Natural Resources
- Forensic Biology
- Psychology
- Commercial Music and Audio Production
- Engineering Technology
- Emergency Management
- Business Administration



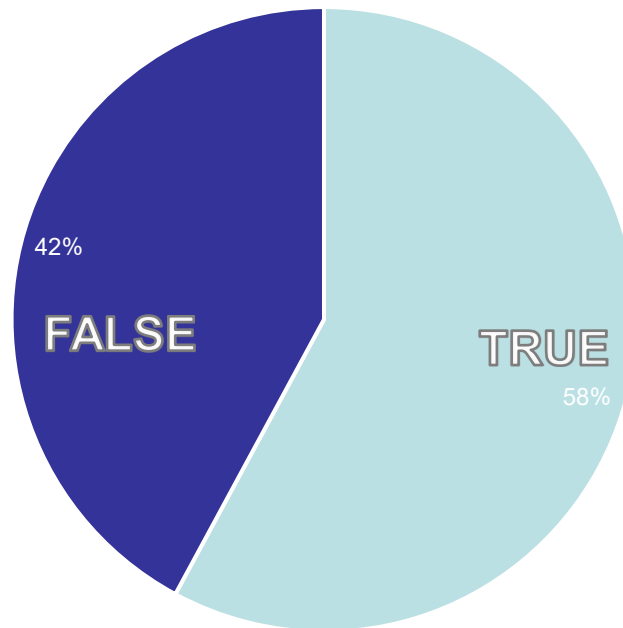
I regularly follow news and updates about AI.

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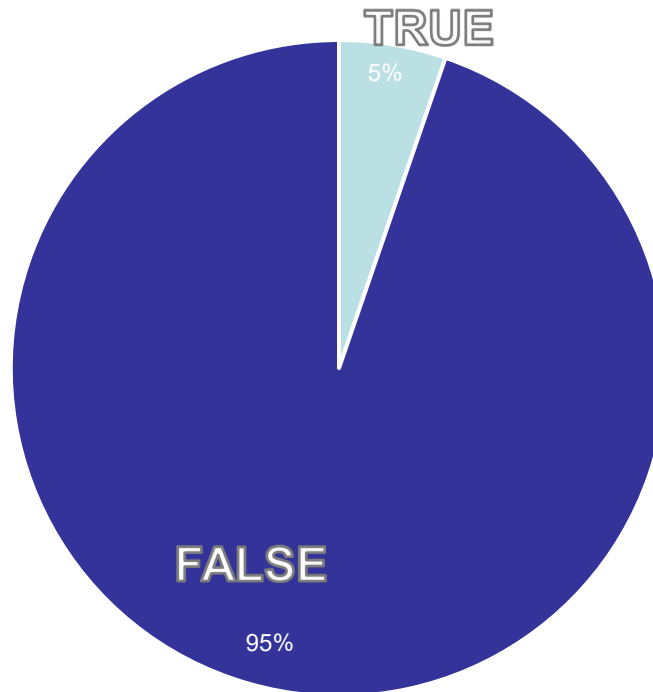
I regularly discuss AI topics with friends, family, or classmates.

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I have attended workshops or seminars on AI.

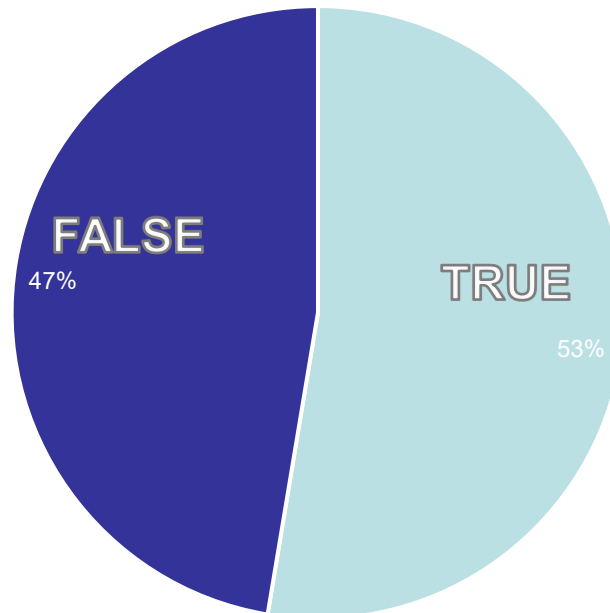
I have attended workshops or seminars on AI.



- This tells us students are not seeking training on their own.
- What does that mean?
- Are they relying on us as faculty?
- Or, are they so tech savvy, they do not need training?

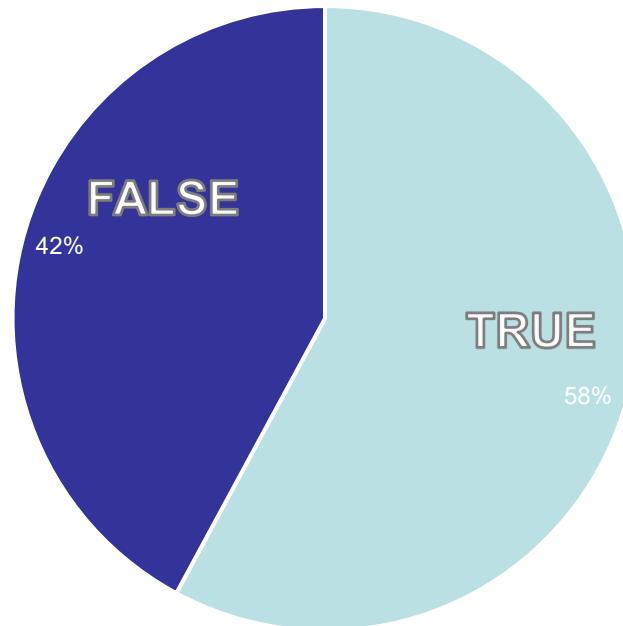
I have seen opportunities to learn more about AI around the WCU campus.

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I regularly use AI-powered tools or applications in my studies.

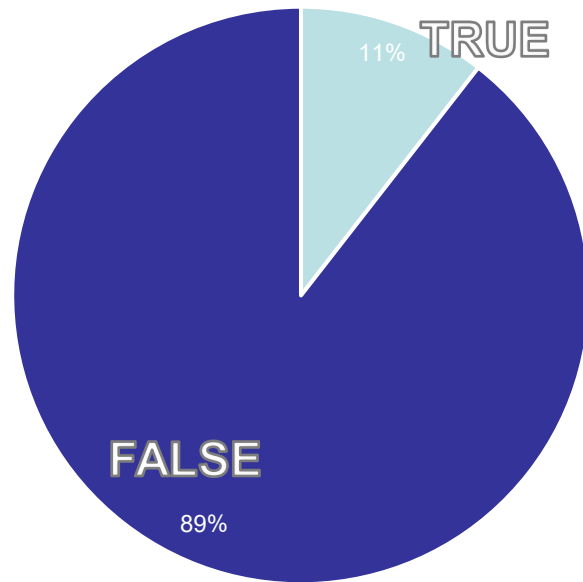
I regularly use AI-powered tools or applications in my studies.



- This tells us students are using a technology in which they may or may not have been properly trained to use.

AI-powered tools are essential for my academic success.

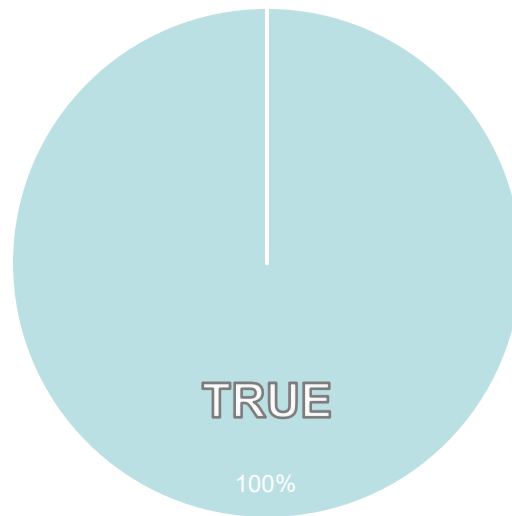
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- Is this a true perception of reality in their future careers?

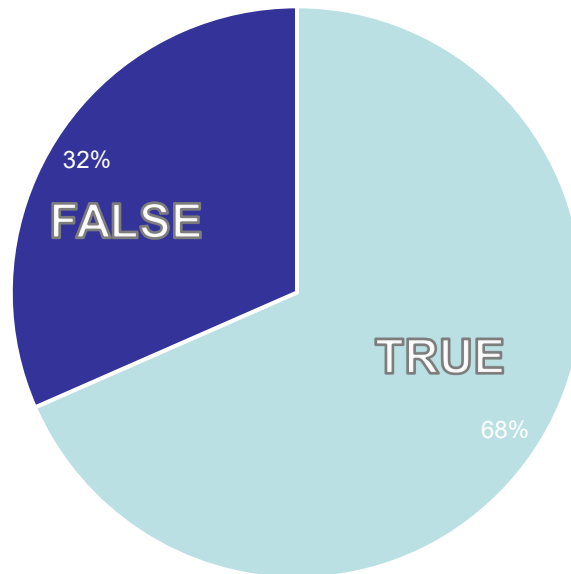
I feel that it is necessary to verify the validity and accuracy of the responses that AI generates.

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AI has positively affected my learning experience at WCU.

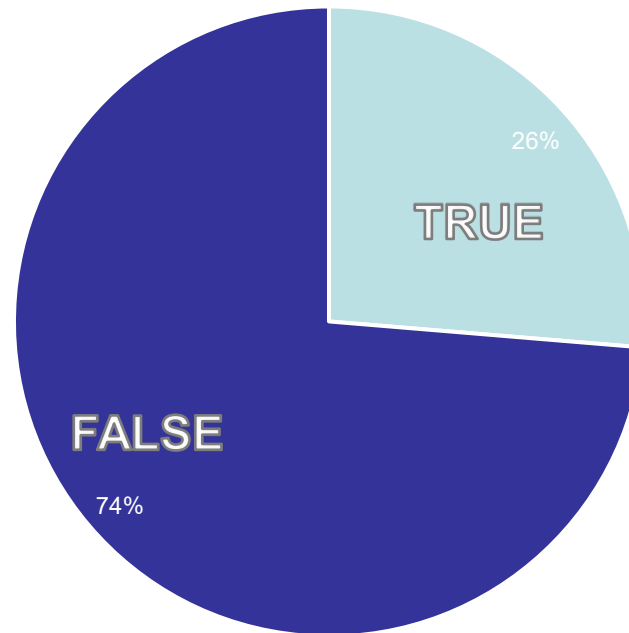
AI has positively affected my learning experience at WCU.



- Students don't feel they need it to learn, but they enjoy the commodity.

AI has negatively affected my learning experience at WCU.

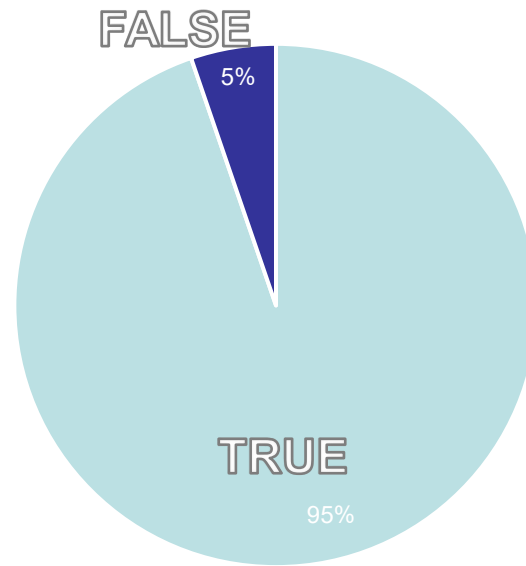
AI has negatively affected my learning experience at WCU.



- We have to remember that AI can have a positive and negative effect at the same time.

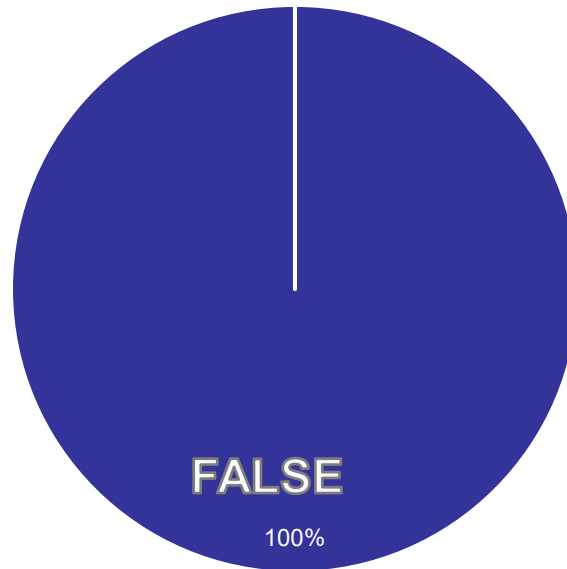
I use AI outside of my classwork.

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I am comfortable submitting a prompt to an AI like ChatGPT and turning in the answer it provides.

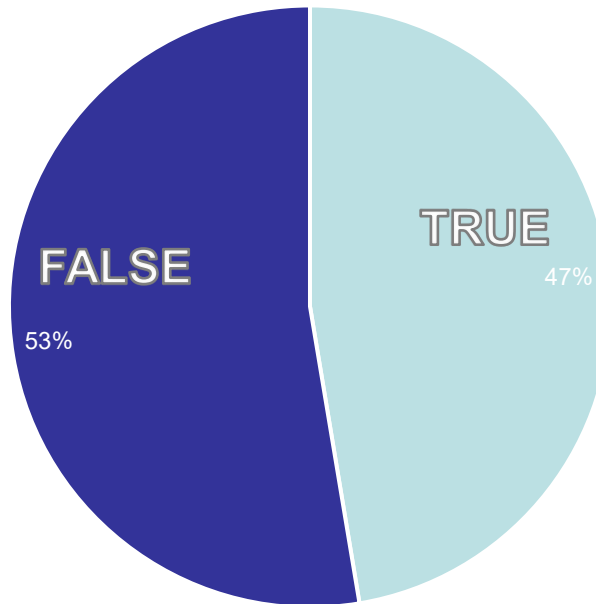
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- This goes along with the prior question and shows consistency in their feelings towards AI.

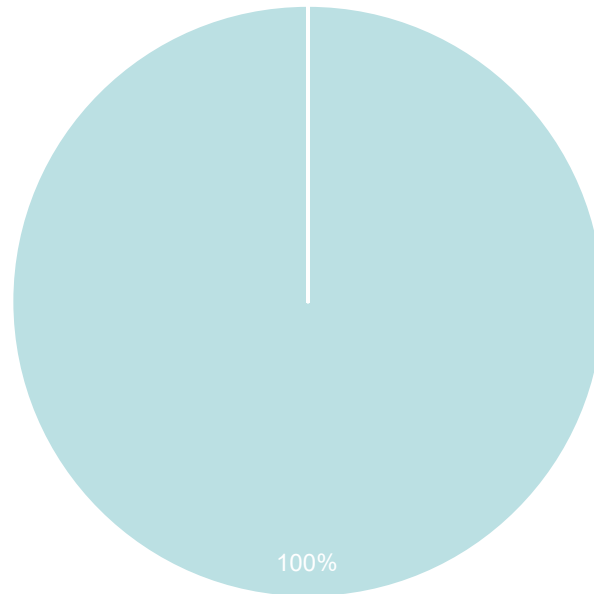
AI can contribute positively to social issues.

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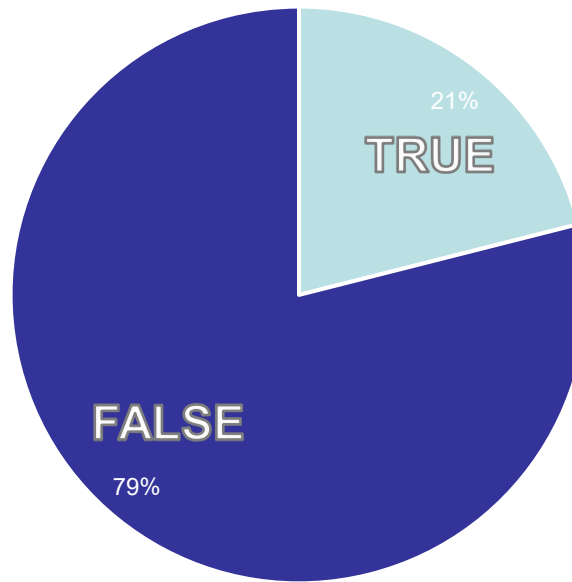
I worry about AI's impact on personal privacy.

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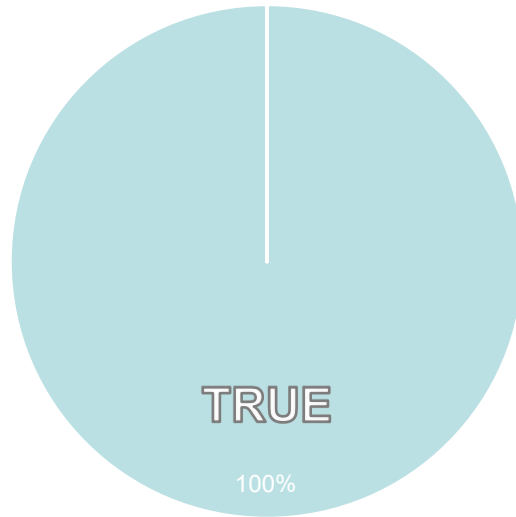
I would feel embarrassed if someone found out that I used AI for schoolwork.

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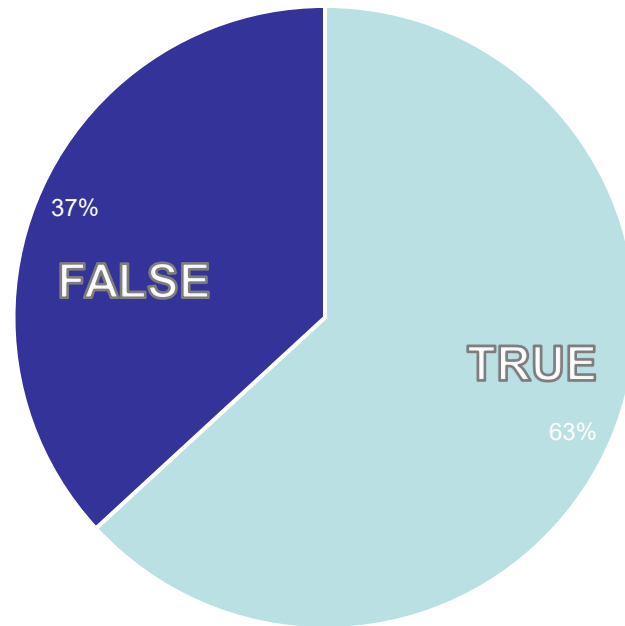
AI algorithms should be more transparent.

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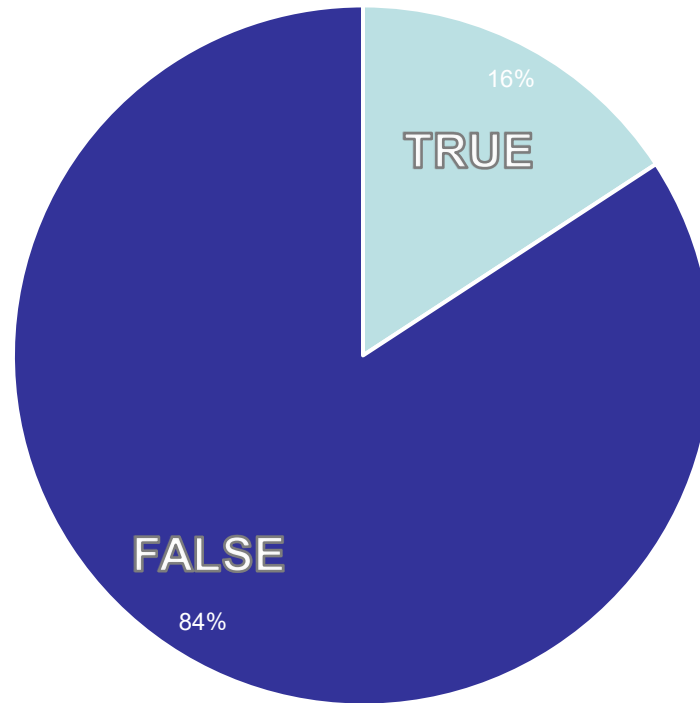
AI technology can enhance creativity and innovation.

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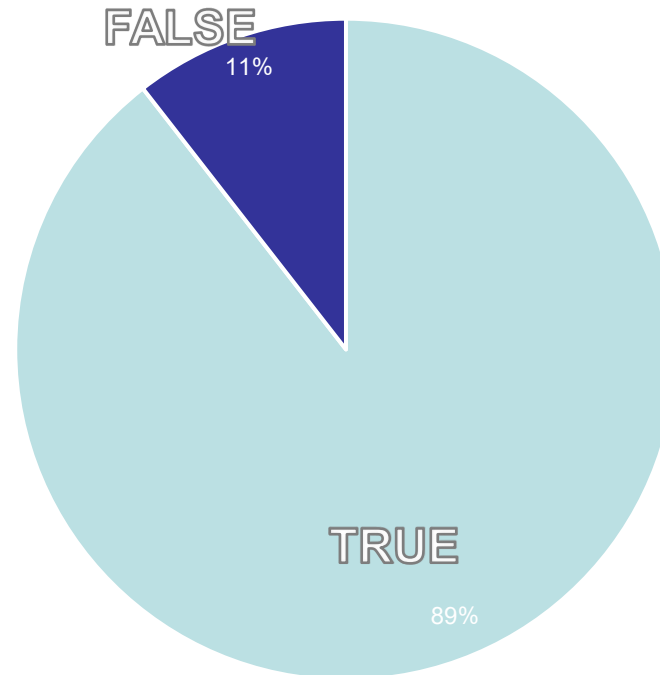
I trust AI algorithms to provide accurate information.

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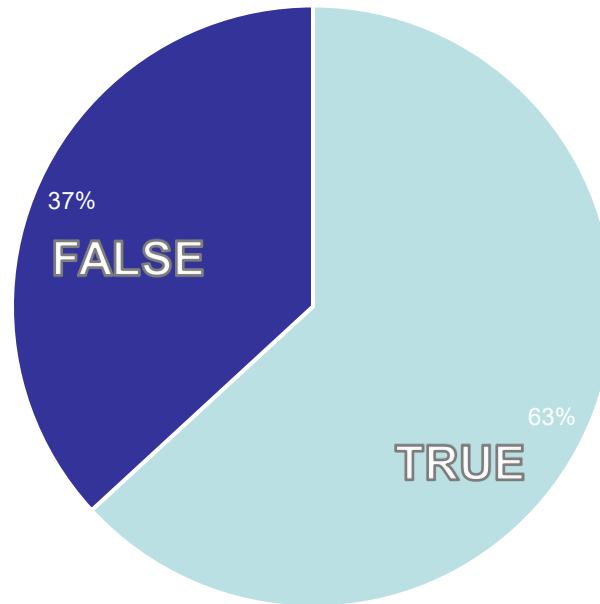
The ethical use of AI is a major concern for me.

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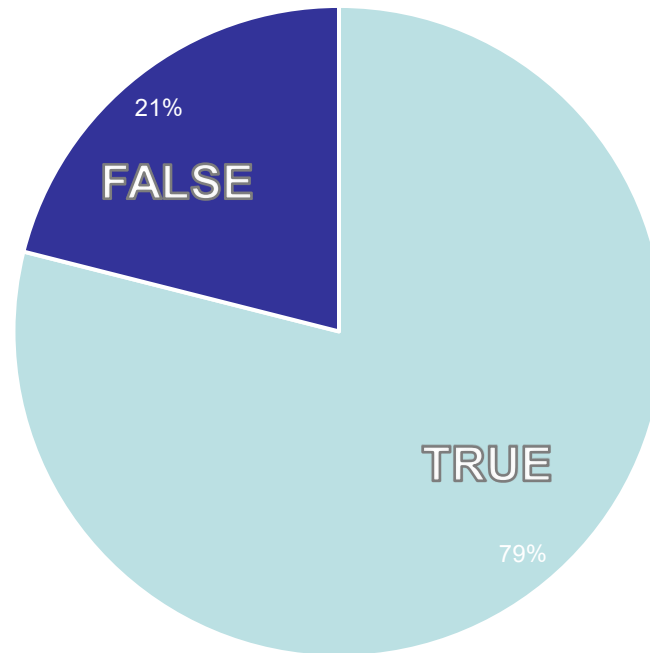
AI has the potential to reduce human biases.

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I have concerns about AI's impact on job security.

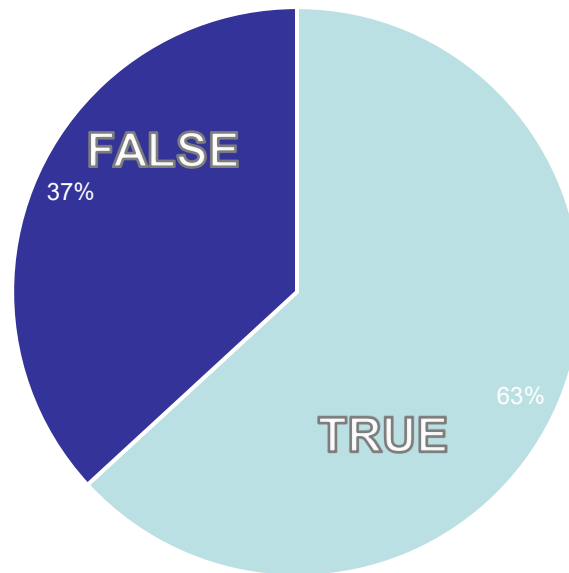
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- Going back to “AI tools are essential for academic success”
- It seems they know AI will have impact, but the weight of impact is uncertain.

I am interested in receiving formal training in AI through coursework or other resources at WCU.

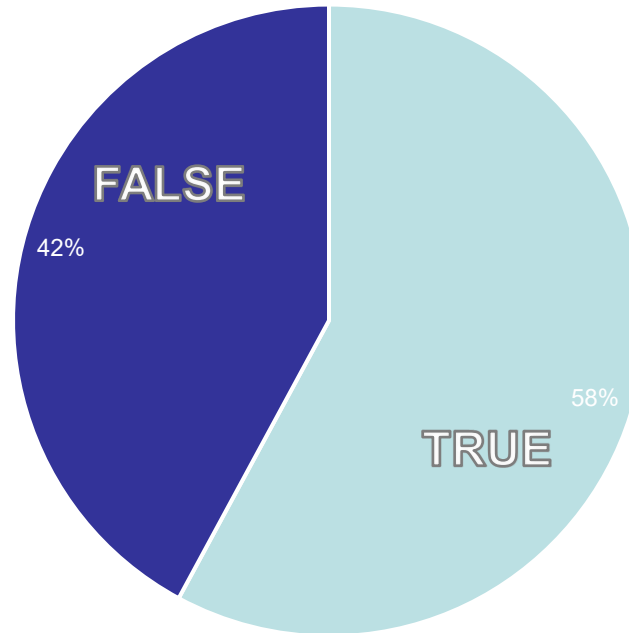
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- Seems they do want to know more.

My curriculum lacks adequate exposure to AI.

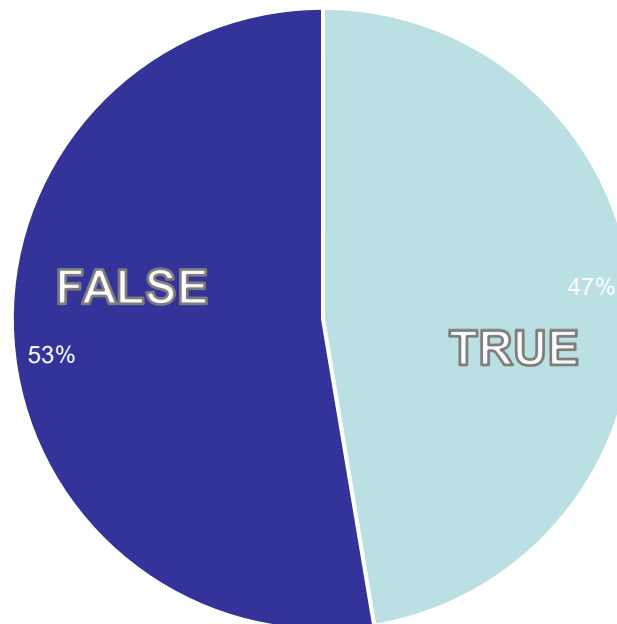
My curriculum lacks adequate exposure to AI.



- As faculty, we have to retool ourselves for the future as well.
- Does the curriculum actually reflect the industry?

Students who use AI for their coursework have an advantage academically.

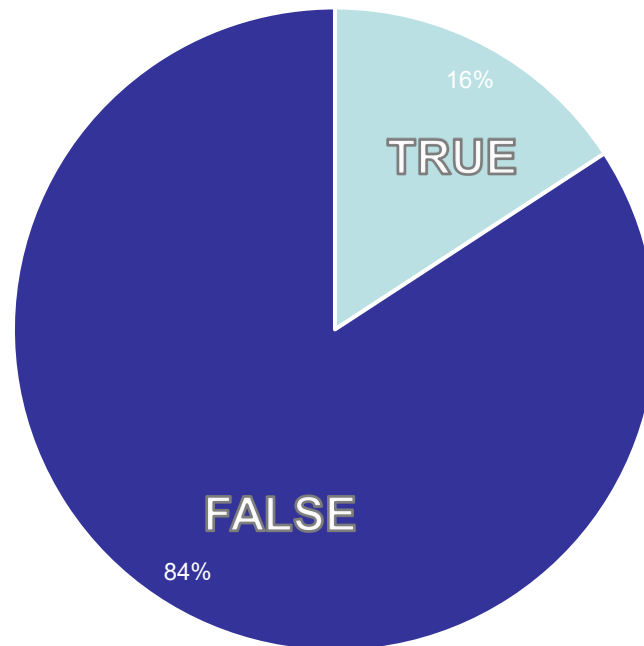
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- Are students feeling slighted?

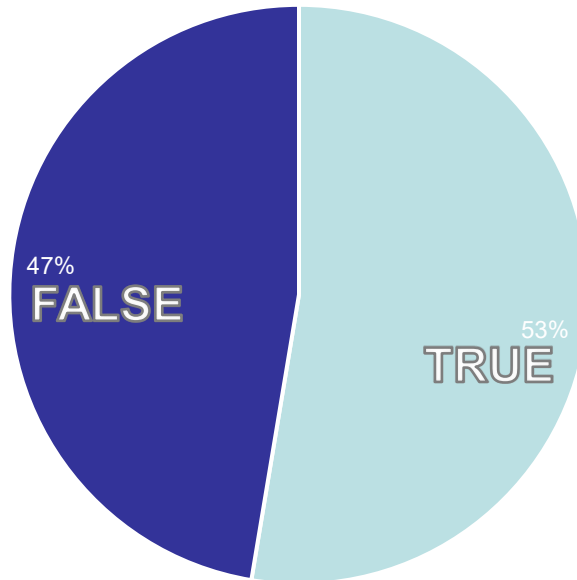
My WCU professors encourage the use of AI in coursework.

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I am actively seeking opportunities to learn more about AI.

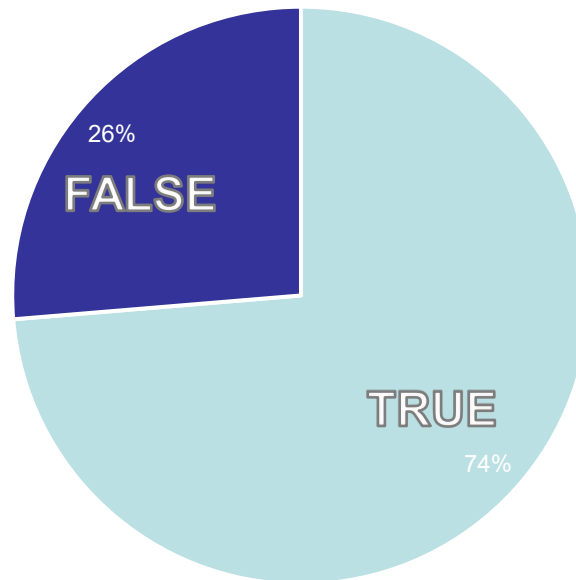
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- We should be providing these opportunities.

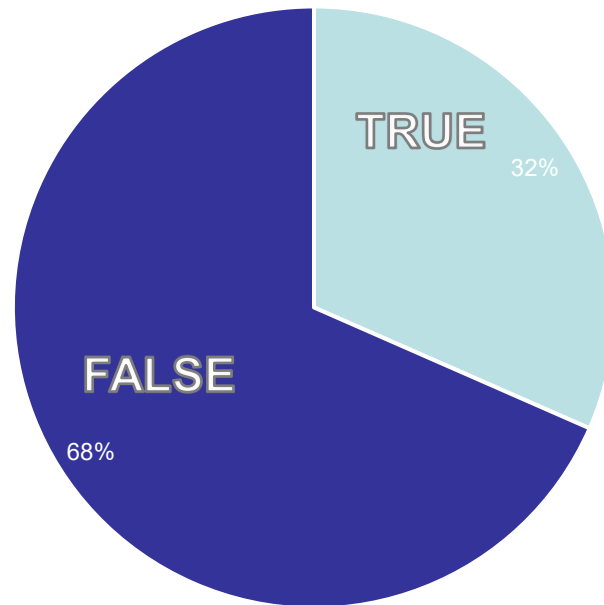
I am skeptical about the benefits of AI in education.

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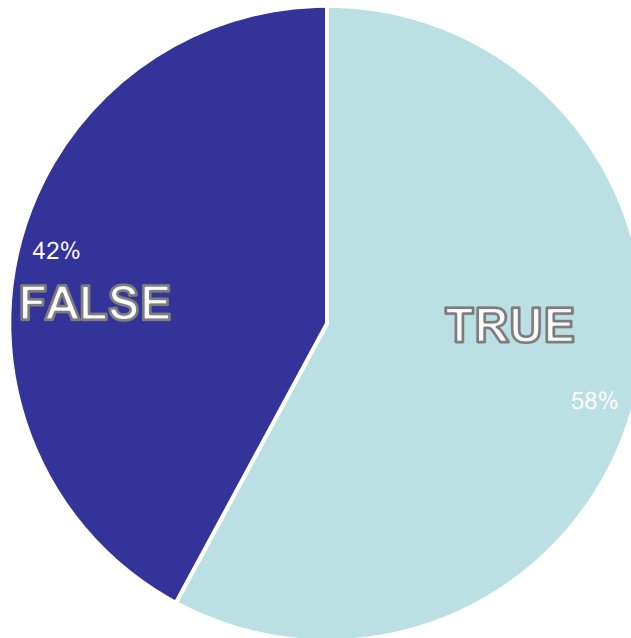
I see potential for AI to solve complex global problems.

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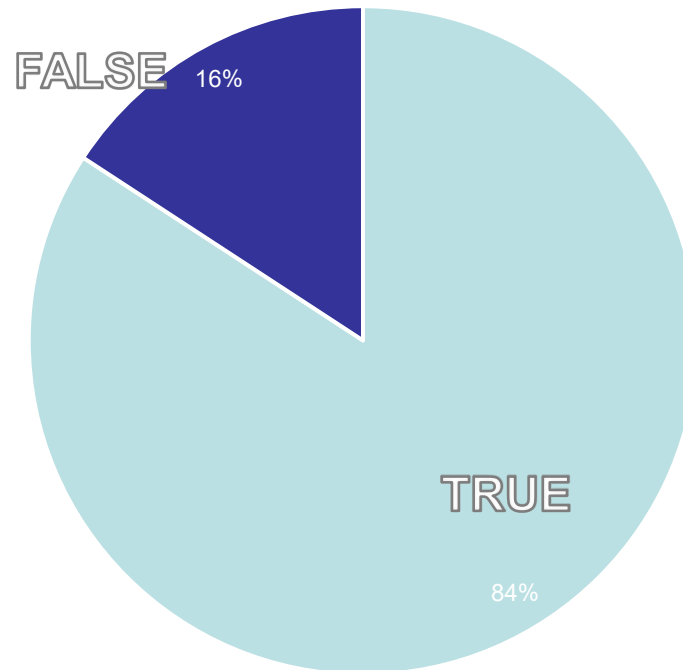
AI will become an essential part of most professions.

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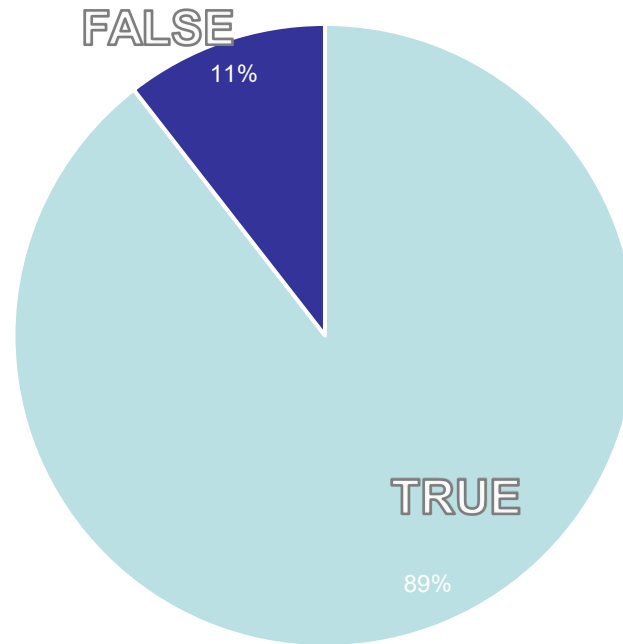
I worry about AI negatively affecting human creativity.

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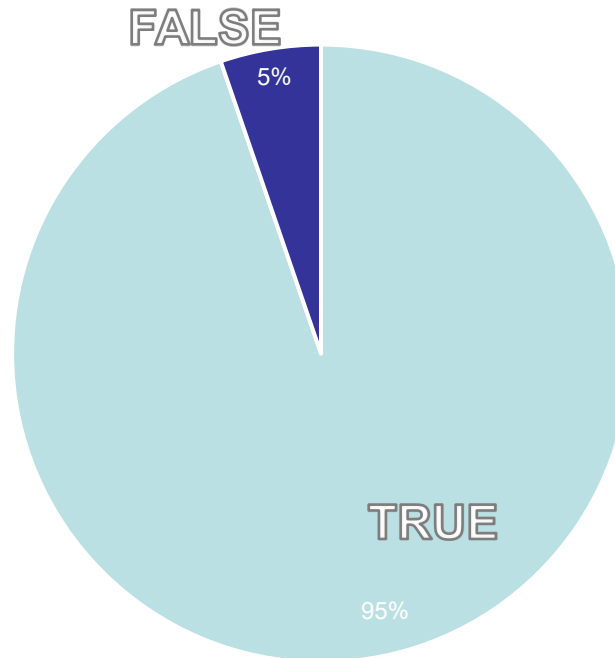
I have concerns about AI's long-term societal impact.

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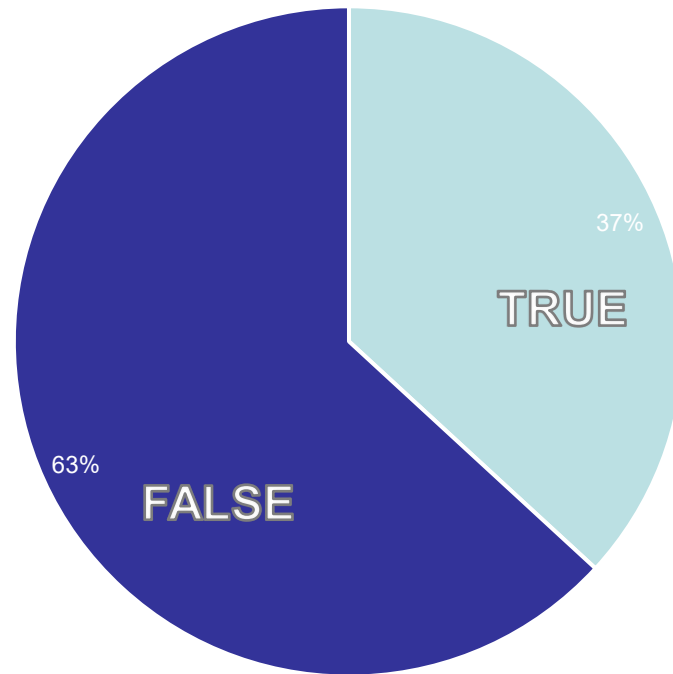
Unregulated AI development may lead to unforeseen risks.

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AI will play a significant role in my future career.

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- What is the role for higher education moving forward?
- In reality, student views on this question are likely skewed.

Goal: Practical strategies for integration.

- **Automates tasks (proofreading, citations, etc...).**
- **Provides instant feedback.**
- **Can spark creativity.**
- **Challenges: plagiarism, over-reliance.**

- **Establish Clear Guidelines**
 - Define acceptable AI use.
 - Sample syllabus language.
 - Discuss ethics and integrity.

Integrating AI Into Assignments

- **Compare student vs. AI drafts.**
- **Use AI for research assistance.**
- **Encourage creative AI applications.**

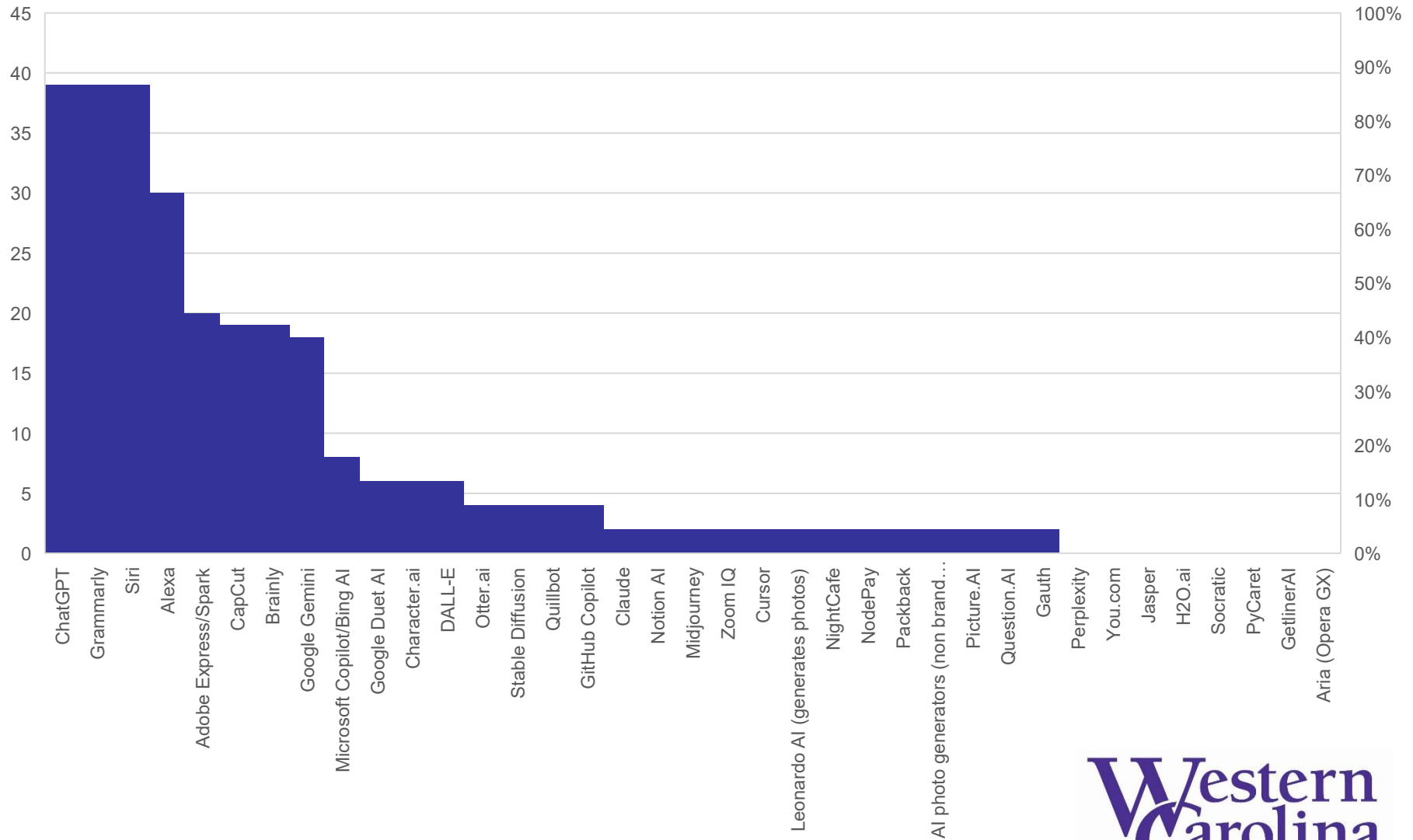
Most popular according to AI

- **ChatGPT (OpenAI)**
- **Grammarly**
- **Google Bard**
- **Microsoft Copilot**
- **Jasper AI**
- **DALL·E / Midjourney (AI Art)**
- **Quillbot**
- **Perplexity AI**
- **Socratic by Google (AI tutoring)**

Survey Results:

ChatGPT
Google Gemini
Microsoft Copilot/Bing AI
Grammarly
Claude
Perplexity
Quillbot
You.com
Brainly
Character.ai
Jasper
DALL-E
Midjourney
Stable Diffusion
Adobe Express/Spark
CapCut
Alexa
Siri
GitHub Copilot
Notion AI
Otter.ai
Google Duet AI
Zoom IQ
Cursor
Leonardo AI (generates photos)
NightCafe
NodePay
Packback
AI photo generators (non brand specific-Instagram, Tiktok)
Picture.AI
H2O.ai
Socratic
Question.AI
Gauth
PyCaret
GetlinerAI
Aria (Opera GX)

Actual Use According to WCU Survey



Teach Critical AI Literacy

How AI works (basics).

- **AI systems learn patterns from large amounts of data.**
- **Examples:**
 - Machine Learning (ML): A branch of AI where algorithms learn from data to make predictions or decisions without being explicitly programmed for each task. (A spam filter learns to recognize spam emails by analyzing thousands of labeled examples.)
 - Neural Networks & Deep Learning: These are more advanced types of ML that mimic how the human brain processes information. They power technologies like image recognition (e.g., face ID) and generative AI (e.g., ChatGPT or DALL·E).
 - Natural Language Processing (NLP): This enables AI to understand and generate human language, which is why tools like ChatGPT can answer questions or draft text.
- **AI systems do not “think”. They find patterns and correlations, but they don’t understand context the same way people do.**

Verify Information

- **AI fact-checking is a major concern.**
- **Generative AI tools (like ChatGPT) can produce convincing answers, but they don't always verify facts.**
- **They generate text based on patterns in their training data. This means they can “hallucinate” (make up facts).**
- **Students and instructors should:**
 - Cross-check answers with credible sources like peer-reviewed journals, reputable news outlets, or textbooks.
 - Use AI tools with built-in citation features when possible and always double-check citations.
 - Understand that AI is only as reliable as the data it was trained on — it can reflect outdated or biased information if not used carefully.
- ***Fact-Checking with AI (Stanford Craft):***
 - <https://craft.stanford.edu/resource/what-is-ai-generated-misinformation/>
- **Others (not endorsing these, just providing options):**
 - <https://toolbox.google.com/factcheck/explorer>
 - <https://mediabiasfactcheck.com/>
 - <https://www.reuters.com/fact-check/>
 - <https://www.invid-project.eu/tools-and-services/invid-verification-plugin/>
 - <https://www.deepware.ai/>

Identify Biases

- **AI is not neutral.**
- **It can reflect or amplify human biases present in its training data.**
 - For example, facial recognition systems have historically shown higher error rates for people of color due to biased datasets (Buolamwini & Gebru, 2018).
- **Students should learn to:**
 - Recognize that AI outputs can reflect cultural, gender, racial, or socioeconomic biases.
 - Ask questions about who built the AI, what data it was trained on, and whose perspectives may be missing.
 - Discuss ethical guidelines for AI development and deployment.

Impact on future careers

World Economic Forum *Future of Jobs Report*

- **What it covers:** The Future of Jobs Report regularly surveys employers worldwide to predict how AI, automation, and other technologies will transform jobs, skills, and industries.
- **Key stat:** The 2023 report estimates that 44% of workers' core skills are expected to change within the next five years due to technological disruption, including AI.
- **Website:** <https://www.weforum.org/reports/the-future-of-jobs-report-2023/>

McKinsey Global Institute *Jobs Lost, Jobs Gained*

- **What it covers:** This report looks at how automation and AI could displace some jobs while creating demand for new ones, especially roles requiring creativity, digital skills, and advanced education.
- **Key stat:** By 2030, up to 375 million workers (14% of the global workforce) may need to switch occupations due to automation.
- **Website:** <https://www.mckinsey.com/featured-insights/future-of-work/jobs-lost-jobs-gained-what-the-future-of-work-will-mean-for-jobs-skills-and-wages>

OECD *AI and the Future of Work*

- **What it covers:** The OECD (Organization for Economic Co-operation and Development) offers detailed country-by-country data on how AI is reshaping work, training, and skills.
- **Key stat:** In OECD countries, 14% of jobs are at high risk of automation, and another 32% will be significantly transformed.
- **Website:** <https://www.oecd.org/employment/Automation-policy-brief-2023.pdf>

Faculty Can...

- **Model Responsible Use**
 - Show how you use AI.
 - Be transparent.
 - Evaluate AI outputs critically.
- **Promote Collaboration & Innovation**
 - Group work with AI.
 - Host workshops/discussions.
 - Invite student suggestions.
- **Address Concerns & Risks**
 - Clear plagiarism policies.
 - Use detection tools wisely.
 - Facilitate open dialogue.

Conclusion

- **AI is here to stay.**
- **Instructors shape wise users.**
- **Embrace AI for engaged, future-ready learning.**
- **Train ourselves, attend workshops, conferences, etc...**
- **Test discipline-relevant tools.**
- **Stay updated, share insights.**
- **How do you see AI in your classroom?**

1.Introduction to AI: [EDUCAUSE AI Reports](#)

2.Understanding AI's Impact: [Turnitin AI Writing Policy](#)

3.Establish Clear Guidelines: [EDUCAUSE Teaching with AI](#)

4.Integrate AI Into Assignments: [UNESCO AI in Education](#)

5.Teach Critical AI Literacy: [Common Sense Education AI Guide](#) | [Google Scholar](#) | [JSTOR](#)

6.Model Responsible Use: [EDUCAUSE AI Toolkit](#)

7.Promote Collaboration & Innovation: [Stanford HAI Student Program](#)

8.Address Concerns & Risks: [FERPA Privacy](#) | [Turnitin AI Detection](#)

9.Resources for Instructors: [ISTE AI for Educators](#) | [OpenAI Educator FAQ](#) | [EDUCAUSE AI Exploratory](#)

10.AI for Distance Learning: [Otter.ai](#) | [Khanmigo](#)

ASEE News-President Signs Executive Orders To Boost US AI Industry

- The [New York Times](#) (7/23, McCabe, Kang) reported in continuing coverage that President Trump “said on Wednesday that he planned to speed the advance of artificial intelligence in the United States, opening the door for companies to develop the technology unfettered from oversight and safeguards, but added that AI needed to be free of ‘partisan bias.’” The Times adds that in a “sweeping effort to put his stamp on the policies governing the fast-growing technology,” the President “signed three executive orders and outlined an ‘AI Action Plan,’ with measures to ‘remove red tape and onerous regulation’ as well as to make it easier for companies to build infrastructure to power AI.”
- [Bloomberg](#) (7/23, Lai, Davalos, Hordern, Subscription Publication) reported that the orders Trump signed “include a measure addressing energy and permitting issues for AI infrastructure, a directive to promote AI exports and one that calls for large language models procured by the government to be neutral and unbiased.” The President said at the event, “America is the country that started the AI race, and as President of the United States, I’m here today to declare that America is going to win it.”
- [Reuters](#) (7/23, Nellis) reported that as part of the effort, the Administration “recommended implementing export controls that would verify the location of advanced artificial intelligence chips, a move that was applauded by US lawmakers from both parties in both houses of Congress.”

AI for Distance Learning

- **AI chatbots for 24/7 student support (e.g., ChatGPT, Bing Chat)**
- **Automated tutoring and feedback tools (e.g., Socratic, Khanmigo)**
- **Virtual teaching assistants (e.g., Google Bard, Microsoft Copilot)**
- **AI-powered discussion moderation**
- **AI transcription & captioning for lectures**
- **Personalized learning pathways with AI analytics**
- **AI tools to detect student engagement in online platforms**

Example Assignments

ChatGPT Discussion Forum

For this discussion, you will need to create your ChatGPT account. The discussion will consist of 4 posts. Follow the steps for each post to fully complete the discussion board. Do not wait until the last day to complete the first post.

First Post

1. Ask the ChatGPT a technical question. For example, "Write a cnc code to 3d print a rabbit" or "What size floor joists should be used on a room that is 210 sq feet?" Come up with a unique question of your own. You can use the OpenAI free version.
2. Share the link for the answer ChatGPT provides. Click the share button in the upper right corner of your chat.

Second Post

3. Choose a link posted from one other classmates' ChatGPT technical examples. Read their questions and ChatGPT answers.
4. Respond to the other student by answering how accurate you found the answer ChatGPT provided to be based on your technical knowledge?

Third Post

5. In your own words what do you think about ChatGPT?
6. What are some common uses you would find important for using ChatGPT?
7. From this experience, what are the limitations of ChatGPT?

Fourth Post

8. Explore at least one other Open Source AI software.
9. Read the reviews. If you want to join and try one, feel free to compare in real time. You do not have to join a second AI platform.
10. Post a comparison review between ChatGPT and the other platform you researched. Topics to discuss in your review can be accuracy, uses, user friendliness, etc...

Teachable Machine

Teachable Machine Discussion Forum

- Teachable Machine is a web-based educational tool developed by Google that allows users to create machine learning models without needing to write any code. It's designed to make machine learning concepts more accessible to a wider audience, including students, educators, and developers who may not have extensive experience in machine learning.
- <https://teachablemachine.withgoogle.com/>
- For this assignment you will read the resources on Teachable Machine to learn more about the machine learning concepts within AI. Use Teachable Machine to create a new project. You can choose any of the three inputs-image, voice, or pose. Develop your project and choose export to create a link to your project.
- First Post: Describe what you did for your project. Include at least two screen shots of your project so other students know what pictures or poses you used. If you use voice, describe what sounds or words you used. Post the link to your exported project.
- Second Post: Look at two other students' projects by clicking on their links. Try it out. Did it work? Were they clear on the prompts for the machine? Post feedback to them on the discussion board.
- Third Post: Try incorporating your code from Teachable Machine into another application. There are two examples posted in Canvas, but you can choose any related application. For your post, describe the application used. Did it work? Did you need more time for development? Post a link if applicable. Post at least two screen shots of your experimentation. I do not expect you to create an amazing app or game here. I just want you to experiment, but if you are successful, please share.

Works Cited

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